

Biology 123 — Master Cheatsheet

Reproduction • Transport • Gas Exchange

CHAPTER 9: REPRODUCTION

- Asexual (4 types)
- Sexual + fertilization
- Plant flowers & seeds

WEEK 10: TRANSPORT

- Animal circulation
- Human heart & vessels
- Plant xylem & phloem

WEEK 11: GAS EXCHANGE

- Fick's Law & surfaces
- Lungs, gills, tracheae
- Stomata & leaves

Big-Picture Framework

The three systems share one logic: life is built on exchanges across surfaces. Reproduction exchanges **genetic information**; transport moves **materials** inside the body; gas exchange moves **gases** between the body and its environment. In every case the rate of the exchange depends on **surface area, concentration gradient, and distance**. Keep this diffusion principle in mind and the entire unit becomes consistent.

Universal Definitions You Will See Repeatedly

- **Diffusion:** Passive movement from high → low concentration.
- **Gradient:** The difference (of pressure, concentration, or water potential) that drives movement.
- **Surface area : volume ratio:** Small organisms rely on diffusion alone; large ones need dedicated transport/exchange systems.
- **Passive vs. active:** Passive = no ATP (diffusion, transpiration). Active = ATP required (loading phloem, ion pumps).
- **Mitosis vs. meiosis:** Mitosis = identical diploid copies (asexual). Meiosis = haploid gametes with recombination (sexual).
- **Haploid (n) / Diploid (2n):** One vs. two sets of chromosomes.
- **Cohesion / Adhesion:** Water-to-water / water-to-walls attraction — drives xylem flow.
- **Fick's Law:** $\text{Rate} \propto (\text{Surface area} \times \Delta\text{Pressure}) / \text{Distance}$.
- **Endotherm vs. ectotherm:** Generates vs. absorbs body heat; endotherms demand more O_2 and food.
- **Source vs. sink (plants):** Source = sugar producer (leaf); sink = sugar user (root, fruit).

Spotlight: Fick's Law of Diffusion

$$\text{Rate} = (A \times \Delta P) / D$$

A = surface area • ΔP = partial-pressure difference across the surface • D = diffusion distance.

- Increase A (folds, alveoli, gill lamellae) → faster exchange.
- Keep ΔP high (ventilate to renew air, circulate blood) → faster exchange.
- Keep D small (one-cell-thick membranes, moist surfaces) → faster exchange.

2 • Animal Asexual Reproduction

Core idea: A single parent produces genetically identical offspring by mitosis. Fast and energy-cheap but produces no variation — vulnerable to environmental change.

The Four Asexual Strategies

A. Fission

- Parent divides into two roughly **equal** parts; each regenerates missing structures.
- **Examples:** planaria (flatworms), sea anemones, single-celled protists.
- Often combined with regeneration — a cut planarian becomes two worms.

B. Budding

- A **small outgrowth** forms on the parent and grows into a new individual.
- The bud is **smaller** than the parent (key difference from fission).
- May detach or remain attached to form a **colony**.
- **Examples:** Hydra, corals, some sponges, yeast (single-celled).

C. Fragmentation

- A piece **breaks off** and regenerates into a full organism.
- Requires strong regenerative tissue; not all animals can do it.
- **Examples:** sea stars (arm + part of central disc), ribbon worms, sponges.
- **Cysts:** fragments too small to regrow form dormant cysts until conditions improve.

D. Parthenogenesis (“virgin birth”)

- Unfertilized egg develops without sperm — offspring may be haploid or diploid.
- **Examples:** Komodo dragons, whiptail lizards (all-female), aphids, New Zealand mud snails, some sharks, honeybees (drones).
- **Honeybees:** fertilized eggs → females; unfertilized → **haploid male** drones.
- **Whiptail lizards:** no males exist, but females still perform courtship to trigger ovulation.

Quick Comparison

Type	Parent piece	Cell basis	Classic example
Fission	Equal halves	Mitosis	Planaria
Budding	Small outgrowth	Mitosis	Hydra
Fragmentation	Any fragment	Mitosis + regeneration	Sea star
Parthenogenesis	Unfertilized egg	Mitosis of egg	Komodo dragon

Asexual vs. Sexual at a Glance

	Asexual	Sexual
Parents	One	Two (usually)
Genetic variation	None (clones)	High (meiosis + recombination)
Speed	Fast	Slower
Energy cost	Low	High (courtship, gametes)
Adaptability	Low	High
Cell division	Mitosis	Meiosis → fertilization

3 • Animal Sexual Reproduction & Fertilization

Core idea: Two parents each contribute **haploid gametes** (sperm and egg) that fuse to form a **diploid zygote**. Produces genetic variation via meiosis.

Adult Sex Determination

Gonochoristic

- Sex is **fixed at birth** — has ovaries OR testes, not both.
- Most vertebrates (humans, dogs, most fish).

Simultaneous Hermaphrodites

- Both functional ovaries and testes **at the same time**.
- Still usually cross-fertilize with a partner (avoids inbreeding).
- **Examples:** earthworms, banana slugs, tapeworms, some fish.

Sequential Hermaphrodites

- **Switch sex** during life — triggered by social/environmental cues.

■ **Protandrous:** Born male → switch to female (e.g., clownfish — dominant one becomes female).

■ **Protogynous:** Born female → switch to male (e.g., blue-headed wrasse, many reef fish).

Embryonic Sex Determination

■ **Genetic:** Chromosomes decide (XX/XY in humans, ZW in birds).

■ **Temperature-sensitive:** Incubation temp sets sex in many turtles/crocodilians — climate change is a threat to these species.

Fertilization Strategies

External Fertilization

- Eggs + sperm released into the environment (usually water).
- Requires **synchronization** (spawning) and **many gametes** to compensate for losses.
- **Examples:** most fish, amphibians, coral (often triggered by lunar/rain cycles).

Internal Fertilization

- Sperm deposited inside the female — higher success, fewer offspring needed.
- Requires courtship and specialized organs.
 - **Spermatophore:** Sperm packet left for the female to pick up (salamanders, some arthropods).
 - **Intermittent organs:** Penis, claspers (sharks), gonopodium (guppies) direct sperm.

Developmental Modes

■ **Oviparous:** Eggs laid externally; embryo develops outside the mother (birds, most reptiles, monotremes).

■ **Oviviparous:** Eggs retained inside; embryo lives on its own **yolk**, no placenta (some sharks, guppies, some snakes).

■ **Viviparous:** Embryo develops inside the mother and is fed **through a placenta** (most mammals, some sharks).

Direct vs. Indirect Development

■ **Direct:** Hatches/born as a miniature adult (grasshoppers, humans).

■ **Indirect:** Distinct larval stage → metamorphosis (caterpillar → butterfly, tadpole → frog).

Mammalian Reproduction & Cycles

- **Monotremes:** Egg-laying mammals (platypus, echidna). Still nurse young with milk.
- **Marsupials:** Short gestation; young born very undeveloped, finish growth in a **pouch** (kangaroos, opossums).
- **Placental (eutherian):** Longest gestation; highly developed at birth; highest parental investment.
- **Estrous cycle:** Most mammals. Female only receptive during **estrus**; unused uterine lining is reabsorbed.
- **Menstrual cycle:** Primates. Unused uterine lining is **shed** (menstruation).
- **Delayed implantation:** Fertilized egg waits to implant so birth aligns with favorable season (some bats, bears).
- **Induced vs. spontaneous ovulation:** Triggered by mating (cats, rabbits) vs. cyclical (humans).
- **r vs. K selection:** r = many offspring, little care (fish, insects). K = few offspring, heavy care (mammals, birds).

4 • Plant Reproduction (Sexual & Asexual)

Angiosperms (flowering plants) dominate this unit. A flower is a reproductive organ that houses both male (stamens) and female (carpels) structures and relies on pollinators or wind for fertilization. Seeds are tiny sporophyte embryos packaged with food in a protective coat.

Flower Anatomy — The Four Whorls

Whorl	Role	Key parts
1. Sepals	Protect the bud	Green, leaf-like (collectively: calyx)
2. Petals	Attract pollinators	Often colored/scented (corolla)
3. Stamens (male)	Produce pollen	Anther (pollen sacs) + filament
4. Carpel (female)	Produce ovules / receive pollen	Stigma, style, ovary (contains ovules)

Plant Sex Arrangements

- **Monoecious:** Male AND female flowers on the same plant (corn, oak).
- **Dioecious:** Male and female flowers on **separate plants** (holly, willow).
- **Perfect flower:** Contains both stamens and carpel.
- **Imperfect flower:** Only one sex.

Pollination

- Transfer of pollen from anther to stigma (**not** fertilization yet).
- Vectors: wind, insects, birds, bats, water.
- Self-pollination vs. cross-pollination trade-off: self = reliable, no variation; cross = variation, needs a vector.

Gametophytes & Double Fertilization

- **Pollen grain:** The **male gametophyte** — contains 2 sperm cells + 1 tube cell.
- **Embryo sac:** The **female gametophyte** inside the ovule — 7 cells, 8 nuclei (1 egg + 2 polar nuclei are the important ones).
- **Pollen tube:** Grows from the stigma down the style, delivering both sperm to the embryo sac.

Double fertilization (unique to angiosperms):

- **Sperm #1 + Egg → Zygote (2n)** — becomes the embryo.
- **Sperm #2 + 2 polar nuclei → Endosperm (3n)** — food reserve for the seed.

Seeds & Fruits

Seed = dormant embryo + food + coat

- **Embryo:** The new sporophyte plant, paused mid-development.
- **Endosperm / cotyledons:** Food tissue (endosperm in monocots; cotyledons absorb it in dicots).
- **Seed coat:** Protects and controls dormancy.
- Dormancy broken by cues: moisture, light, fire, gut passage.

Fruit = mature ovary

- **Pericarp:** Ovary wall — **exocarp** (skin), **mesocarp** (flesh), **endocarp** (innermost, often stony).
- Function: protect seeds and **aid dispersal** (animals eat fruit, seeds pass intact).
- Fleshy fruits: apples, berries. Dry fruits: nuts, grains, legumes.
- Seedless fruits come from chemical treatment or ploidy manipulation (e.g., triploid bananas).

Plant Asexual (Vegetative) Reproduction

- **Apomixis:** Diploid megaspore mother cell divides by **mitosis** (skipping meiosis) — produces a clonal seed.
- **Runners / stolons:** Horizontal stems above ground (strawberries).
- **Rhizomes:** Horizontal underground stems (ginger, iris, bamboo).
- **Tubers:** Swollen underground stem portions (potatoes).
- **Suckers:** New shoots from the root (raspberry, aspen groves).
- **Adventitious plantlets:** Tiny clones along leaf margins (Mother of Thousands, kalanchoe).

Why plants do both: vegetative reproduction rapidly fills a good habitat; seeds provide variation and long-range dispersal.

5 • Animal Transport Systems

Why animals need transport: as body size grows, diffusion alone can't move O_2 , nutrients, wastes, and hormones far enough or fast enough. A **circulatory system** bulk-moves fluid to shorten the diffusion distance every cell experiences.

Fundamentals & Pre-Circulatory Strategies

- **Transport system:** Network that distributes substances (water, nutrients, O_2 , hormones) and removes wastes.
- **Diffusion only:** Used by microscopic life (amoebas) and very small invertebrates — works because distances are tiny.
- **Environmental water:** Sponges and cnidarians pump **seawater** through a gastrovascular cavity as a stand-in circulatory system.

Open vs. Closed

- **Open:** Hemolymph is pumped into channels then **bathes organs directly** before draining back. Insects, most arthropods, most mollusks.
- **Closed:** Blood stays entirely inside **vessels**; a heart pumps. Vertebrates, annelids, cephalopods (octopus/squid).
- Closed systems support higher metabolic rates and finer regulation.

Vertebrate Heart Variation

Group	Chambers	Circulation	Notes
Fish	2 (1 atrium, 1 ventricle)	Single	Blood through heart once per lap; oxygenated at gills.
Amphibians	3 (2 atria, 1 ventricle)	Double (with mixing)	Can shunt blood past lungs while submerged.
Most reptiles	3 (partial septum)	Double (with mixing)	Shunting possible; crocodiles nearly 4-chambered.
Birds & Mammals	4	Double, fully separated	No mixing. Supports endothermy & high O_2 demand.

Double Circulation in Detail

- **Pulmonary circuit:** Right side of heart → lungs → back to left side. Picks up O_2 , drops CO_2 .
- **Systemic circuit:** Left side of heart → body → back to right side. Delivers O_2 , picks up CO_2 .
- **Advantage:** Pressure can be much higher in the systemic loop than in the delicate lungs — two pumps, two pressures.
- **Mnemonic:** Left side handles **L**arge-body pressure; Right side handles **R**espiratory (lung) pressure.

Blood Vessels & Blood Components

Vessels

- **Arteries:** Thick, muscular, elastic walls. Carry blood **away** from the heart (usually oxygenated).
- **Veins:** Thinner walls, larger lumen, have **valves**. Return blood **to** the heart.
- **Capillaries:** One-cell-thick walls — the actual exchange surface between blood and tissues.

Blood

- **Plasma:** Liquid component — water, ions, proteins, hormones.
- **Erythrocytes (RBCs):** Carry O_2 via hemoglobin.
- **Leukocytes (WBCs):** Immune defense.
- **Platelets:** Clotting.

6 • Human Transport System

The Cardiac Cycle & Its Pacemaker

Cycle Phases

- **Systole:** Muscle **contraction** — blood is squeezed out.
- **Diastole:** Muscle **relaxation** — chambers fill.
- Atrial systole fills ventricles; ventricular systole ejects blood to lungs/body.

Electrical Conduction Path

- **SA node** (right atrium) → atria contract.
- → **AV node** pauses signal ~0.1 s so atria empty first.
- → Bundle of His → Purkinje fibers → ventricles contract from the apex upward.

Key Terms

- **SA node:** The **pacemaker** — fires ~60–100 times/min in adults.
- **AV node:** Delays the signal; acts as a backup pacemaker.
- **ECG/EKG:** Graph of this electrical activity (P wave = atria; QRS = ventricles; T wave = ventricle reset).
- **Stroke volume:** Blood ejected per beat (~70 mL).
- **Cardiac output:** Heart rate × stroke volume ≈ **5 L/min** — roughly all of your blood passes through every minute.

Blood Pressure & Its Regulation

- **Blood pressure:** Force of blood pushing outward on vessel walls. Reported as **systolic / diastolic** (e.g., 120/80 mm Hg).
- **Hypertension:** Chronic elevation ($\geq 140/90$ typically). Major risk for stroke, heart attack, kidney damage.
- **Hypotension:** Abnormally low — dizziness, fainting.
- **Baroreceptors:** Pressure sensors (aorta, carotids) that trigger adjustments via nervous + endocrine systems (e.g., epinephrine).
- **Vasoconstriction / vasodilation:** Smooth muscle in arterioles narrows or widens vessels to raise/lower pressure and redirect flow.

Lymphatic System & Gas Carriage

Lymphatic System

- Capillaries are **leaky** — fluid oozes into interstitial spaces.
- Lymphatic vessels collect this fluid and return it to the veins near the heart.
- Also transports dietary fats (chylomicrons) and immune cells.
- **Edema:** Swelling caused by failure of this return system.

How Blood Carries Gases

- **O₂:** ~98% bound to **hemoglobin** on RBCs; rest dissolved in plasma.
- **CO₂:** ~70% carried as bicarbonate (HCO₃⁻) in plasma, some bound to hemoglobin, some dissolved.
- **Bohr effect:** Low pH / high CO₂ / high heat → hemoglobin **releases** O₂ — so active tissues get more.
- **Myoglobin:** Muscle-only O₂-binding pigment; extra reserve in diving mammals and working muscle.

Respiratory Pigments Across the Animal Kingdom

Pigment	Metal	Color	Where found
Hemoglobin	Iron	Red	Most vertebrates; inside RBCs
Hemocyanin	Copper	Blue	Mollusks & arthropods (dissolved in hemolymph)
Hemerythrin	Iron	Violet/pink	Some marine worms
Chlorocruorin	Iron	Green	Some marine annelids

7 • Plant Transport & Stress Responses

Core idea: Plants have two vascular tissues — **xylem** moves water/minerals up from the soil, and **phloem** moves sugars wherever they're needed. Most movement is passive, powered by evaporation and osmotic gradients.

Vascular Tissues

	Xylem	Phloem
Transports	Water + dissolved minerals	Sugars (sucrose), amino acids, signals
Direction	One-way: up only	Bidirectional: source → sink
Cell state at maturity	Dead (hollow tubes)	Living
Conducting cells	Tracheids (narrow, tapered); Vessels (wider, stacked)	Sieve-tube elements (no nucleus)
Support cells	—	Companion cells (ATP to load sugars)
Driven by	Transpiration pull (passive)	Pressure-flow (active loading)

Mechanisms of Movement

Transpiration – Cohesion – Tension

- Water evaporates from **stomata** in leaves → negative pressure (tension).
- **Cohesion** (water-to-water H-bonds) + **adhesion** (water-to-xylem-walls) keep the water column intact.
- The whole column is pulled up like a straw — all the way from root to leaf.

Water Potential (ψ)

- Combines **solute potential** (concentration) and **pressure potential**.
- Water moves from high ψ to low ψ .
 - **Turgid:** Cell is swollen — normal, healthy state.
 - **Plasmolysis:** Cell shrivels away from wall from water loss (hypertonic conditions).

Pressure-Flow in Phloem

- **Source** (leaf) actively loads sucrose into sieve tubes → water follows by osmosis → **high pressure**.
- **Sink** (root, fruit, growing tip) unloads sucrose → water leaves → **low pressure**.
- Sugar flows from high to low pressure. Sources and sinks can switch seasonally.

Root Uptake

- **Endodermis:** Inner root layer — the “bouncer.”
- **Casparian strip:** Wax (suberin) band that forces water/ions to pass through cell membranes (selective filtering) instead of around cells.

Plant Stress Responses

Drought

- **Cavitation:** Air bubble breaks the xylem water column — flow stops in that vessel.
- **Stomatal pits & trichomes (hairs):** Shield stomata from drying wind; reduce transpiration.
- **CAM & C_4 pathways:** Alternative photosynthesis that minimizes water loss (CAM opens stomata only at night).

Flooding

- **Lenticels:** Spongy pores in bark for gas exchange when roots are oxygen-starved.
- **Aerenchyma:** Spongy air-filled tissue that pipes O_2 from leaves down to submerged roots.
- **Pneumatophores:** “Knee” roots that grow above water (mangroves) to get O_2 .
- **Adventitious roots:** Roots that form from non-root tissue in response to stress.

8 • Gas Exchange Fundamentals & Animal Systems

Key Vocabulary

- **Gas exchange:** Passive diffusion of gases (usually O_2 in, CO_2 out) across a moist surface.
- **Respiration (biological):** Building ATP. Three stages:
 - **External** — ventilation/gas exchange with environment.
 - **Internal** — gas transport in blood.
 - **Cellular** — ATP production inside mitochondria.
- **Aqueous process:** Gases must dissolve in fluid to diffuse — exchange surfaces must be moist.
- **Partial pressure (P):** Pressure contributed by one gas in a mixture — diffusion chases differences in partial pressure.

Ventilation Mechanisms

- **Buccal pumping:** Mouth/throat muscles gulp air or water (fish, amphibians).
- **Ram ventilation:** Swim with the mouth open so water flows past the gills (many sharks, tuna).
- **Costal ventilation:** Rib/intercostal muscles expand the chest → negative pressure draws air in (reptiles, mammals).
- **Tidal ventilation:** Air flows in and out the **same path** — mammalian breathing.
- **Countercurrent flow:** Blood in gill capillaries flows **opposite** to water — keeps ΔP high across the entire gill (fish).
- **Cross-current flow:** Bird lung design — one-way airflow so air is always fresh.

Animal Gas Exchange Structures

Group	Structure	Key features
Aquatic (fish)	Gills	Lamellae increase surface area; operculum covers gills in bony fish; countercurrent blood/water flow.
Amphibians	Lungs + skin	Thin, moist, vascularized skin enables cutaneous respiration .
Insects	Tracheal system	Spiracles (closable openings) → branching tracheae → tracheoles contact cells directly. No lungs, no blood O_2 .
Birds	Lungs + air sacs	Parabronchi allow one-way flow; cross-current exchange; efficient at altitude.
Mammals	Lungs	Bronchi → bronchioles → alveoli (huge SA, thin, moist). Tidal ventilation.

Why Countercurrent Flow Is a Big Deal

If **blood and water flowed in the same direction**, they'd equalize — blood could grab only half the O_2 . **Because they flow opposite**, blood always meets water that's a little richer in O_2 than itself, **maintaining ΔP along the entire gill length**. Fish can extract 80–90% of dissolved O_2 — a direct application of Fick's Law.

Human Breathing Controls

- Rate is primarily set by **CO_2 / pH** sensors, *not* O_2 . Holding your breath fails because CO_2 rises.
- Medulla oblongata sends signals to the **diaphragm** and **intercostals**.
- Bohr effect (see page 6) makes unloading at working muscles automatic.

9 • Plant Gas Exchange, Leaves & Study Summary

Plants need CO_2 in (for photosynthesis) and O_2 in (for cellular respiration at night). They also make O_2 during the day. All exchange happens passively through pores — mainly **stomata** on leaves and **lenticels** on woody stems.

Stomata, Guard Cells & Leaf Tissues

- **Stomata:** Microscopic pores (usually on the underside of leaves) for gas exchange.
- **Guard cells:** Paired cells flanking each stoma; open/close it based on **turgor pressure** (water + K^+ ions).
 - Daytime, water-abundant: turgid → open for CO_2 .
 - Heat/drought: flaccid → closed to save water.
- **Cuticle:** Waxy layer over epidermis that **blocks** water loss (and gas diffusion) — forces exchange through stomata.
- **Lenticels:** Spongy pores on stems/trunks of woody plants — allow gas exchange through bark.

Mesophyll

- **Palisade mesophyll:** Tightly packed, column-like cells near upper surface — main site of photosynthesis.
- **Spongy mesophyll:** Loose cells with **air spaces** that let gases diffuse to every cell.

CAM Photosynthesis

- Stomata open at **night**, store CO_2 as malic acid.
- Stomata **closed** during the day to conserve water; stored CO_2 fuels photosynthesis.
- **Examples:** cacti, pineapples, many succulents.

Leaf Morphology & Modifications

Part / Type	Description
Blade	Wide, flat photosynthetic surface of the leaf.
Petiole	Stalk joining blade to stem.
Stipules	Small outgrowths at the base of the petiole in some plants.
Simple leaf	Blade undivided (may still have lobes or teeth).
Compound leaf	Blade divided into leaflets (e.g., walnut).
Bracts	"Floral leaves" that look like petals (poinsettia's red "petals").
Spines	Leaves reduced to protective points (cacti) — also reduces water loss.
Tendrils	Thin curling leaves used for climbing (peas).
Insectivorous leaves	Trap and digest insects in nutrient-poor habitats (Venus flytrap, sundew, pitcher plants).

Unit Summary — Shared Themes to Remember

- **Exchange follows Fick's Law:** Every system in this unit (lungs, gills, capillaries, stomata, leaves) maximizes surface area, maintains a gradient, and minimizes distance.
- **Water is the universal solvent:** Gas exchange needs moist surfaces; transport needs water columns; fertilization often happens in water.
- **Energy cost drives strategy:** Asexual is cheap but inflexible; sexual is expensive but adaptive. Open circulation is cheap; closed is powerful. Diffusion is free; active transport costs ATP.
- **Closed & closed — then open:** Internal fertilization, closed circulation, and stomata all solve the same problem: protect a delicate exchange from the environment while still allowing exchange to happen.
- **Variation is the endpoint:** Whether through meiosis (animals), double fertilization (plants), or long-distance seed dispersal, sexual strategies exist because a varied population survives change.